

Inline and Display Mathematics

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\arg \max_{x' \in \mathbb{R}^d} \left[\sum_{1 \leq \ell \leq 2} \varphi_{\ell}(x', \lambda_{\ell}) \right] = \left(\sum_{1 \leq \ell \leq 2} \lambda_{\ell} V_{\ell, XX}^{-1} \right)^{-1} \left(\sum_{1 \leq \ell \leq 2} a_{\ell} V_{\ell, XX}^{-1} V_{\ell, XY} \right),$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\bar{G}^{(\zeta)}(\beta) = \frac{(\beta + i\pi) \sinh\left(\frac{\pi\beta}{2\pi + \zeta}\right)}{\Gamma\left(1 - \frac{\pi - i\beta}{2\pi + \zeta}\right) \Gamma\left(1 - \frac{\pi + i\beta}{2\pi + \zeta}\right)},$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\underline{D}_{\mu}^{\psi}(\eta) \asymp_{\times} \min_{p \in P_{<\delta}} \liminf_{\xi \in G(p)} \left(\frac{1/\|\xi - \eta\|}{1/r_{\xi}} \psi_p(\|\xi - \eta\|) \right)^{k_p - \delta} = \min_{p \in P_{<\delta}} \left(\limsup_{\xi \in G(p)} \frac{r_{\xi}}{\theta_p(\|\xi - \eta\|)} \right)^{k_p - \delta}.$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\sup_{\mathcal{T} > 0} \left| \frac{1}{\mathcal{T}} \int_0^{\mathcal{T}} h(u(t), y_{\epsilon}(t), z_{\epsilon}(t)) dt - \frac{1}{\mathcal{T}} \int_0^{\mathcal{T}} \tilde{h}(\mu(t), z(t)) dt \right| \leq \alpha_h(\epsilon), \quad \text{where} \quad \lim_{\epsilon \rightarrow 0} \alpha_h(\epsilon) = 0$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\gamma^{\mu\rho\nu}\nabla_{\rho}(\Psi_{\nu}^{\mp})^A-\gamma^{\mu\nu}\gamma^5\partial_5(\Psi_{\nu}^{\pm})^A-k\epsilon\tanh\left(k|y|-\frac{k\pi r_c}{2}\right)\gamma^{\mu\nu}\gamma^{\hat{5}}(\Psi_{\nu}^{\pm})^A-\frac{3}{2}k(\sigma_1)^A{}_B\gamma^{\mu\nu}(\Psi_{\nu}^{\mp})^B=0\;.$$